

Boosting Protein–Ligand Binding Pose Prediction and Virtual Screening Based on Residue–Atom Distance Likelihood Potential and Graph Transformer

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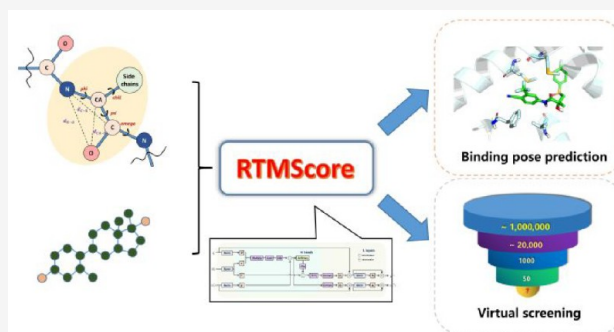


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ABSTRACT: The past few years have witnessed enormous progress toward applying machine learning approaches to the development of protein–ligand scoring functions. However, the robust performance and wide applicability of scoring functions remain a big challenge for increasing the success rate of docking-based virtual screening. Herein, a novel scoring function named RTMScore was developed by introducing a tailored residue-based graph representation strategy and several graph transformer layers for the learning of protein and ligand representations, followed by a mixture density network to obtain residue–atom distance likelihood potential. Our approach was resolutely validated on the CASF-2016 benchmark, and the results indicate that RTMScore can outperform almost all of the other state-of-the-art methods in terms of both the docking and screening powers. Further evaluation confirms the robustness of our approach that can not only retain its docking power on cross-docked poses but also achieve improved performance as a rescoring tool in larger-scale virtual screening.



INTRODUCTION

The accurate prediction of protein–ligand binding modes is a long-standing challenge in structure-based drug design. Several experimental techniques such as X-ray diffraction,^{1,2} nuclear magnetic resonance (NMR) crystallography,³ and cryo-electron microscopy (EM)⁴ can be used to determine the structural details of protein–ligand complexes; however, they are always expensive, and their availability is difficult for all desired cases. Molecular docking has been proposed as a computational alternative, which has posed a considerable impact on drug design, including hit/lead discovery and the following lead optimization.^{5–8} Typically, a docking program needs a searching engine to sample the possible binding poses of the studied molecule in the binding pocket first and then uses a scoring function (SF) to assess their binding strength, and the top-ranked pose is usually considered as the most reasonable binding conformation. Despite impressive achievements in the past several decades, the inaccuracy of SFs continues as a major bottleneck in reliable molecular docking for real-world applications.^{9,10}

The SFs are typically divided into four types, namely, physics-based SFs, empirical SFs, knowledge-based SFs, and machine-learning-based SFs (MLSFs).¹¹ The former two are usually described as the weighted sum of several physical/empirical energy terms/descriptors. Knowledge-based SFs are built through the statistical analysis of the distribution of some

geometric features in protein–ligand structures and can then also be represented as the sum of all pairwise statistical potentials between proteins and ligands. The rapid development of machine learning (ML) and artificial intelligence (AI) in recent years leads to the emergence of MLSFs, most of which can directly learn the function form from data while not relying on the predefined additive formulated hypothesis.^{12–15} However, despite the substantially superior performance as compared to the former three classical SFs in most studies, MLSFs have suffered from a cloud of doubts due to their poor generalization capability.^{16–21} For example, Gabel et al.¹⁶ found that their MLSFs trained on random forest (RF) and support vector machine (SVM) could reproduce the claimed excellent scoring power (the capability for binding affinity prediction) of RFscore,²² but their docking power (the capability for the discrimination of near-native poses from decoy poses) and screening power (the capability for the discrimination of active ligands from decoy compounds) are rather bad. Our previous

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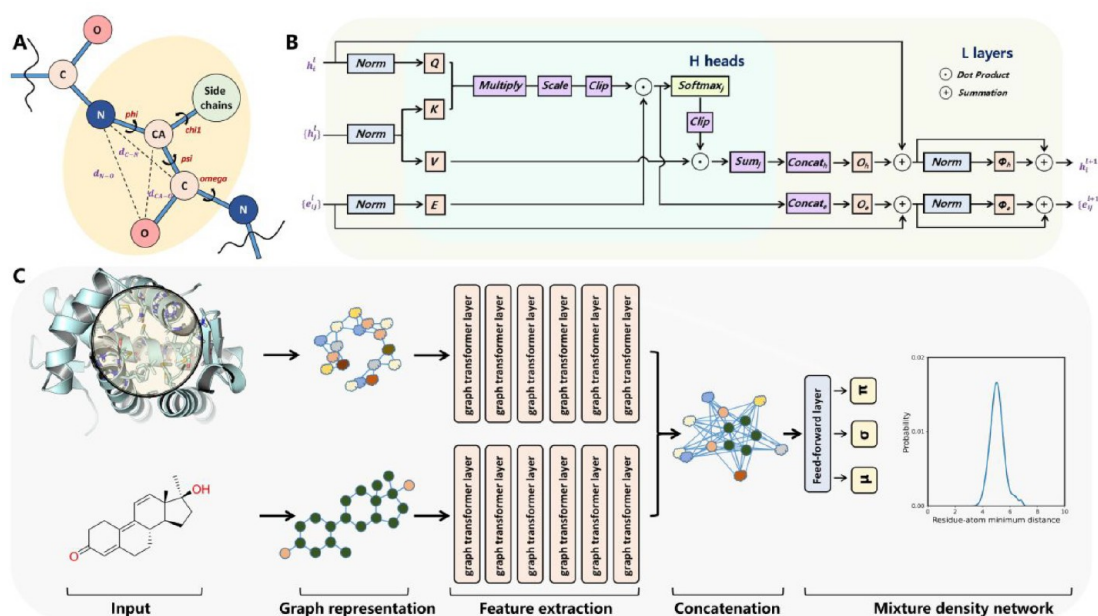


Figure 1. Schematic view of (A) the representation of each residue, (B) the graph transformer layer, and (C) the overall model architecture of RTMScore. The protein and ligand graphs are first processed by two groups of independent graph transformer layers, and then, the extracted features are concatenated in a pairwise manner and input into a mixture density network. The final statistical potential can be obtained through the summation of the probability distributions of all node-pairwise distances.

assessment toward 14 existing MLSFs reaches a similar conclusion: most MLSFs majorly trained on complexes in the PDBbind data set with known experimental data performed remarkably worse than the classical Glide SP on several larger test sets in terms of virtual screening (VS) capability.¹⁷ Hence, considering the inherent essence of ML models, most current MLSFs are task-specific for binding affinity prediction,^{14,22–24} binding pose prediction,^{25,26} and/or VS.^{12,18,27–29} Typically, the models trained on pure experimentally determined protein–ligand complexes may be just applicable to scoring, while those tailored for docking or screening need extra decoy poses or inactive compounds as negative samples for model construction. Nevertheless, classical SFs trained just based on the PDBbind data set can be directly generalized to other tasks.

Despite this, extensive efforts have been made to develop MLSFs with a wider application domain during the past few years.^{30–36} In 2017 and 2019, Zhang's group successively developed $\Delta_{\text{Vina}}\text{RF}_{20}$ ³⁰ and $\Delta_{\text{Vina}}\text{XGB}$,³¹ which employ ML algorithms to fit correction terms to AutoDock Vina scores rather than conventionally fit the final binding scores. These two methods could rank the top three in all four tasks (i.e., scoring, docking, ranking, and screening) in the Comparative Assessment of Scoring Functions (CASF) benchmark, although their VS performance in our previous assessment was unsatisfactory.¹⁷ Recently, they further proposed $\Delta_{\text{Lin_F9}}\text{XGB}$,³² which employed a similar parametrization strategy based on another empirical SF Lin_F9. This new SF could not only achieve excellent scoring, ranking, and screening powers on the CASF benchmark but also outperform the baselines in terms of the large-scale docking-based VS on the LIT-PCBA data set. OnionNet-SFCT proposed by Zheng et al.³³ also incorporated a correction term to AutoDock Vina, but this term was calculated based on a binding pose prediction model rather than binding affinity prediction, implying that the training of the model still relies on the involvement of decoy poses. The linear combination of Vina scores and root-mean-square error

(RMSD) values produced by the ML model enabled the method to perform quite well in terms of docking and screening powers. The emergence of deep learning (DL) has brought in several new strategies for the development of MLSFs with higher generalization ability. A feasible strategy is to build a multitask neural network to simultaneously handle the scoring, docking, and screening tasks, and a striking example is MT-Net.³⁴ Another impressive strategy is employed in PIGNet,³⁵ which calculates the atom–atom pairwise interactions based on physics-informed equations parametrized with neural networks and then sums these interactions to obtain the total binding affinity. This strategy retains the additive function form of empirical SFs and just uses neural networks to fit the interactions at the atom level, thus significantly improving the model generalization. Even so, both of these strategies still need additional decoys for data argumentation, making them far from the original definition of SF. Very recently, Méndez-Lucio et al.³⁶ proposed a geometric DL-based method named DeepDock that creatively introduced a mixture density network (MDN) to learn the probability density distribution of the distance between each ligand atom and each point in the surface of the binding site. The negative log-likelihood of all pairs could be integrated as a statistical potential, which was similar with the construction of most knowledge-based SFs. This distance likelihood potential performed competitively in docking and screening tasks, indicating that it was a promising strategy for capturing protein–ligand interactions. However, the proteins were all triangulated into the form of surface meshes and then converted to mesh graphs with relative Cartesian coordinates as edges, thus making the method sensitive to the rotation of proteins. In addition, with the rapid advances in graph representation learning, there is still space to further develop MLSFs based on distance likelihood potentials.

Herein, we reported RTMScore, a novel SF based on residue–atom prediction likelihood potential and graph transformer, for the prediction of protein–ligand interactions. The

Table 1. Node and Edge Features Employed for Ligand Graph Construction

features	size	description
Nodes		
atom_type_one_hot	17	one hot encoding for atom type ([“C”, “N”, “O”, “S”, “F”, “P”, “Cl”, “Br”, “I”, “B”, “Si”, “Fe”, “Zn”, “Cu”, “Mn”, “Mo”, “other”])
atom_degree_one_hot	7	one hot encoding for atom degree ([0, 1, 2, 3, 4, 5, 6])
atom_formal_charge	1	formal charge
atom_num_radical_electrons	1	number of radical electrons
atom_hybridization_one_hot	6	one hot encoding for atom hybridization ([“sp”, “sp2”, “sp3”, “sp3d”, “sp3d2”, “other”])
atom_is_aromatic	1	whether the atom is aromatic
atom_total_num_H_one_hot	5	one hot encoding for total number of Hs on the atom ([0, 1, 2, 3, 4])
atom_chirality_one_hot	3	one hot encoding for chirality of an atom ([“R”, “S”, “other”])
Edges		
bond_type_one_hot	4	one hot encoding for bond type ([“SINGLE”, “DOUBLE”, “TRIPLE”, “AROMATIC”])
bond_is_conjugated	1	whether the bond is conjugated
bond_is_in_ring	1	whether the bond is in a ring
bond_stereo_one_hot	4	one hot encoding for the stereo configuration of a bond ([“STEREONONE”, “STEREOANY”, “STEREOZ”, “STEREOE”])

Table 2. Node and Edge Features Employed for Protein Graph Construction

features	size	description
Nodes		
residue_type_one_hot	32	one hot encoding for residue type ([“GLY”, “ALA”, “VAL”, “LEU”, “ILE”, “PRO”, “PHE”, “TYR”, “TRP”, “SER”, “THR”, “CYS”, “MET”, “ASN”, “GLN”, “ASP”, “GLU”, “LYS”, “ARG”, “HIS”, “MSE”, “CSO”, “PTR”, “TPO”, “KCX”, “CSD”, “SEP”, “MLY”, “PCA”, “LLP”, “metal”, “other”])
residue_self_distance	5	maximum and minimum values of the scaled distance (multiplied by 0.1) within any atom in a residue, and the scaled distance (multiplied by 0.1) between the atoms named CA and O, the distance (multiplied by 0.1) between the atoms named O and N, and the distance (multiplied by 0.1) between the atoms named C and N
residue_dihedral_angle	4	scaled dihedral angles (multiplied by 0.01), including <i>phi</i> , <i>psi</i> , <i>omega</i> , and <i>chi1</i>
Edges		
residue_is_connected	1	whether two residues are covalently connected
residue_CA_distance	1	scaled distance (multiplied by 0.1) between the CA atoms of two residues
residue_center_distance	1	scaled distance (multiplied by 0.1) between the center of two residues
residue_maximum_distance	2	maximum and minimum values of the scaled distance (multiplied by 0.1) between two residues

proteins and ligands were first characterized as three-dimensional (3D) residue graphs and two-dimensional (2D) molecular graphs, respectively, followed by two groups of independent graph transformer layers to learn the node representations of proteins and ligands. Then, all node features were concatenated in a pairwise manner and input into an MDN to calculate the parameters needed for a mixture density model. Through this model, the probability distribution of the minimum distance between each residue and each ligand atom could be obtained and aggregated into a statistical potential by summing all independent negative log-likelihood values. Compared with DeepDock, the major innovations of RTMScore include the following: (1) representing the proteins as undirect graphs at the residue level; and (2) using graph transformer for the feature extraction of proteins and ligands. These changes enable our method to achieve a state-of-the-art performance on the CASF-2016 benchmark in terms of both the docking and screening powers. Further investigation on extra data sets confirms that RTMScore is a robust SF that possesses an excellent generalization capability. Moreover, our approach can estimate the contributions at the atom or residue level, highlighting the good interpretability of the method.

RESULTS AND DISCUSSION

RTMScore Achieves State-of-the-Art Docking and Screening Powers on the CASF-2016 Benchmark. The performance of RTMScore (Figure 1 and Tables 1 and 2) was first assessed on the standard CASF-2016³⁷ benchmark and compared with several popular traditional SFs. The scoring

power in terms of Pearson correlation coefficient (R_p) and the ranking power in terms of Spearman correlation coefficient (R_s) can be found in Figure S1, and the primary indicators regarding the docking and screening powers are shown in Figure 2. We also collected the results relevant with the docking and screening powers of some recently developed MLSFs and summarize them in Table 3. For the sake of comparison, we also trained a fork of DeepDock (denoted as DeepDockM) on our data set, as well as several models by combining our residue-based graph representation for proteins and model architecture adopted in DeepDock (denoted as RTMScoreM), in order to explore if graph transformer consistently works for the extraction of node representations.

As expected, due to the underutilization of binding affinity data, RTMScore cannot yield satisfactory scoring ($R_p = 0.455 \pm 0.015$) and ranking ($R_s = 0.529 \pm 0.004$) powers, and the methods with a similar strategy such as RTMScoreM ($R_p = 0.489 \pm 0.016$ and $R_s = 0.531 \pm 0.039$) and DeepDockM ($R_p = 0.499 \pm 0.025$ and $R_s = 0.469 \pm 0.040$) also perform badly. However, our approach can achieve state-of-the-art docking and screening powers in terms of almost all of the metrics. With respect to the docking power, if excluding the crystal poses in the test sets, in an average of 93.4% of cases, the top-ranked pose scored by our method is within the predefined RMSD ≤ 2.0 Å of the native binding poses, and this value increases to 97.3% if the crystal poses are retained; when the top-three poses are considered, the two success rates (SRs) can even increase to 98.9% and 99.6%, respectively. RTMScoreM and DeepDockM can yield the average top1 success rates of 89.7% and 87.9%, respectively, without the consideration of the native poses, and

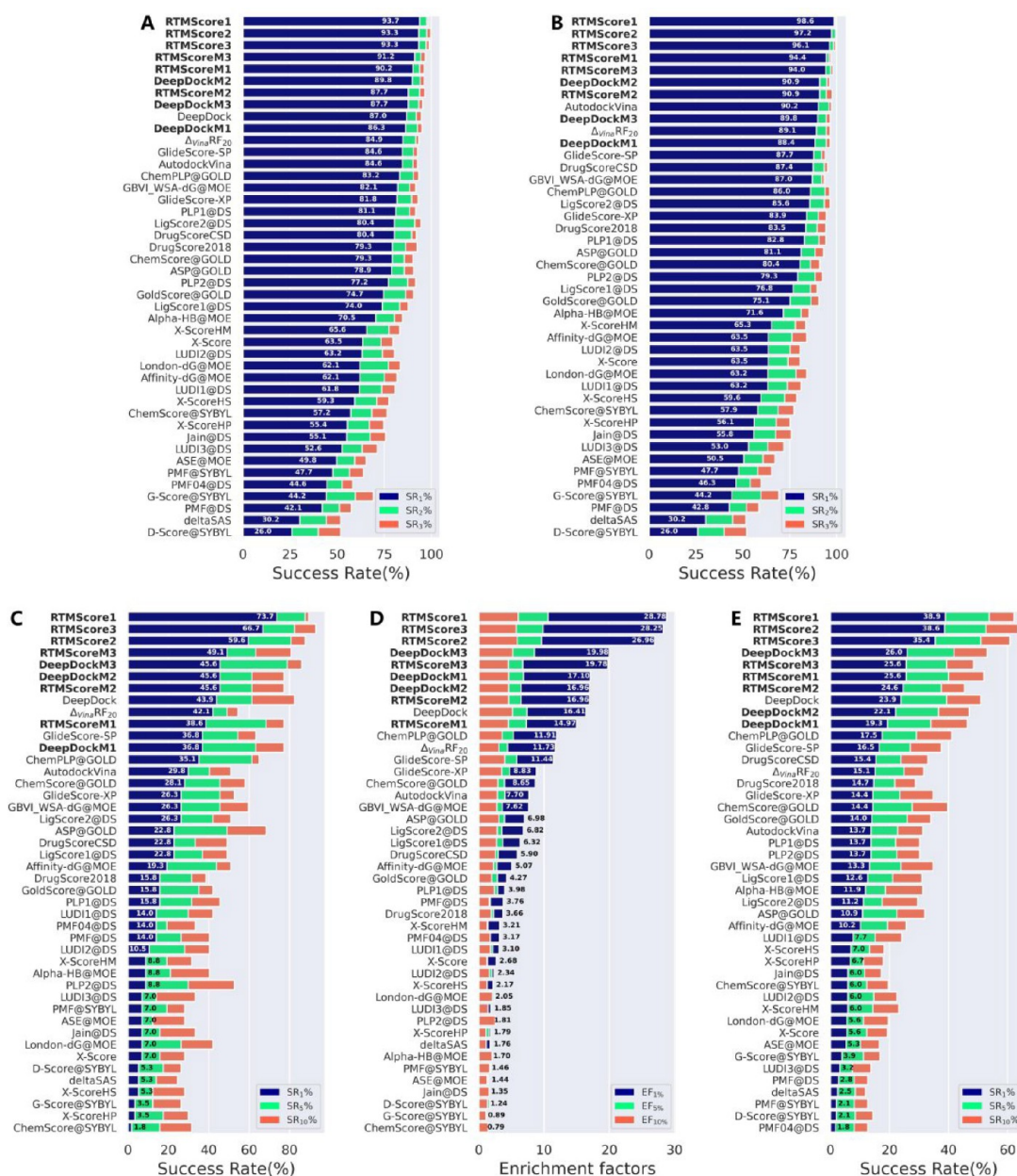


Figure 2. Performances of scoring functions on the CASF-16 benchmark, including the docking powers in terms of (A) success rates excluding crystal poses and (B) success rates including crystal poses, and screening powers in terms of (C) success rates and (D) enrichment factors in the forward screening and (E) success rates in the reverse screening. The models retrained in this study are in bold font. All of the methods are ranked in a descending order in terms of (A, B, C, and E) top1 success rates and (D) EF_{1%}.

turn to 93.1% and 89.7% with the consideration, suggesting that our residue-based graph representation and graph transformer-based feature extractor can be indeed beneficial to the improvement of the docking power. When it comes to the results of binding funnel analysis (Figure S2), the gaps between the methods based on distance likelihood (including RTMScore, RTMScoreM, DeepDockM, and DeepDock) are not so obvious, but all of them exhibit significantly higher R_s than the other methods. This high rank correlation may be one of the major driving forces for the high success rates of these types of methods.

As for the screening power, RTMScore still exhibits its overwhelming superiority over other approaches. In the “forward screening” test (Figure 2C,D and Table 3), among all 57 targets, RTMScore can identify the highest-affinity binder

for an average of 38 (66.7%), 47.7 (83.6%), and 51.3 (90.1%) targets when the 1%, 5%, and 10% top-ranked ligands are considered, even superior to PIGNet³⁵ (SR_{1%} = 55.4%), a physics-informed DL model that shows excellent generalization capability but includes a lot of decoys for model training. In addition, it can achieve average EF_{1%}, EF_{5%}, and EF_{10%} values of 28.00, 10.02, and 5.90, respectively, which are comparable with KORP-PL⁴¹ (EF_{1%} = 22.23), a coarse-grained knowledge-based SF, but still prominently higher than other compared SFs. In the “reverse screening” test (Figure 2E), among all 285 ligands, RTMScore can find the true target for an average of 107.3 (37.6%), 149.3 (52.4%), and 177 (62.1%) ligands when the 1%, 5%, and 10% top-ranked targets are considered, while the second-ranked DeepDock can just obtain the corresponding success rates of 23.9%, 39.3%, and 50.9%. Integrating all three

Table 3. Comparison of RTMScore with Other State-of-the-Art Methods in Terms of Docking Power and Forward Screening Power on the CASF-2016 Benchmark

scoring function	docking power		forward screening power	
	top1 success rate (without native poses)	top1 success rate (with native poses)	top1 success rate	EF _{1%}
Autodock Vina ³⁸	0.846	0.902	0.298	7.70
ChemPLP@GOLD ³⁹	0.832	0.860	0.351	11.91
GlideScore-SP ⁴⁰	0.846	0.877	0.368	11.44
KORP-PL ⁴¹	0.856	0.891	0.421	22.23
$\Delta_{\text{Vina}}\text{RF}_{20}$ ³⁰	0.849	0.891	0.421 (0.456)	11.73 (12.36)
$\Delta_{\text{Vina}}\text{XGB}$ ³¹		0.916	0.368	13.14
$\Delta_{\text{Lin_F9}}\text{XGB}$ ³²		0.867	0.404	12.61
OnionNet-SFCT +Vina ³³		0.937	0.421	15.50
DeepBSP ²⁵	0.872 ± 0.014	0.885 ± 0.013		
PIGNet ³⁵	0.870		0.554	19.6
DeepDock ³⁶	0.870		0.439	16.41
DeepDockM	0.879 ± 0.018	0.897 ± 0.012	0.427 ± 0.508	18.01 ± 1.70
RTMScoreM	0.897 ± 0.018	0.931 ± 0.019	0.444 ± 0.535	17.24 ± 2.42
RTMScore	0.934 ± 0.002	0.973 ± 0.013	0.667 ± 0.071	28.00 ± 0.94

^aThe values in the brackets are reported by the Zhang group,³¹ while the values outside of the brackets are copied from the original paper of the CASF-2016 benchmark.³⁷

metrics of the screening power into account, RTMScoreM (SR_{1%} = 44.4% ± 5.3%, EF_{1%} = 17.24 ± 2.42, and SR_{1%} = 25.3% ± 0.6%) does not necessarily outperform DeepDockM (SR_{1%} = 42.7% ± 5.1%, EF_{1%} = 18.01 ± 1.70, and SR_{1%} = 22.5% ± 3.4%), but the dominant performance of RTMScore still verifies the feasibility of our strategy for screening.

RTMScore Retains Its Superior Docking Power on Cross-Docked Poses. Considering that the assessment of the docking power on the CASF benchmark is only based on redocked poses, which can hardly exist in real-world scenarios, we further tested our trained models on the cross-docking data sets proposed in our previous study (PDBbind-CrossDocked-Core)²⁶ to investigate if they can be generalized to cross-docked poses. These data sets were built based on the PDBbind-v2016 core set and were generated with three docking programs (i.e., Surflex-Dock,⁴² Glide SP,⁴⁰ and AutoDock Vina³⁸) by docking the ligands in the same cluster into the pocket(s) of the corresponding protein (redocked poses) or the other proteins (cross-docked poses), respectively, thus leading to a total of 6 subsets. As shown in Table 4, our models can consistently outperform the traditional methods (i.e., the corresponding docking methods, Vina scores extracted from NNScore⁴³ features, X-Score,⁴⁴ and Prime-MM/GBSA⁴⁵). For example, RTMScore could successfully identify 81.0% and 50.8% of the redocked and cross-docked complexes, respectively, if the top-ranked poses predicted by Surflex-Dock are considered, while the best-performed classical methods could only obtain the corresponding success rates of 75.4% and 43.8%.

In the previous study, we tried to construct several ML classifiers based on either the redocked or cross-docked poses yielded by Surflex-Dock to directly distinguish near-native poses from incorrect poses. We found that ML models trained on cross-docked poses performed better than the counterpart

Table 4. top1 Success Rates of Scoring Functions on PDBbind-CrossDocked-Core Data Set, Where the Poses Are Generated by Surflex-Dock, Glide SP, or AutoDock Vina

method ^a	Surflex		Glide		Vina	
	re ^c	cross	re	cross	re	cross
Docking	0.709	0.438	0.656	0.422	0.551	0.318
Vina	0.695	0.398	0.599	0.359	0.554	0.321
X-Score	0.670	0.418	0.582	0.362	0.512	0.328
Prime-MM/GBSA	0.754	0.427	0.681	0.410	0.460	0.225
NNScore_XGB_Re	0.768	0.479	0.642	0.376	0.505	0.284
NNScore+Rank_XGB_Re	0.800	0.478	0.663	0.421	0.561	0.336
ECIF+Vina_XGB_Re	0.737	0.475	0.638	0.414	0.502	0.312
ECIF+Vina+Rank_XGB_Re	0.789	0.501	0.716	0.460	0.614	0.361
NNScore_XGB_Cross	0.744	0.521	0.663	0.445	0.533	0.352
NNScore+Rank_XGB_Cross	0.765	0.517	0.677	0.444	0.551	0.352
ECIF+Vina_XGB_Cross	0.754	0.504	0.663	0.453	0.495	0.348
ECIF+Vina+Rank_XGB_Cross	0.744	0.522	0.667	0.471	0.526	0.355
DeepDockM1	0.793	0.480	0.723	0.450	0.653	0.374
DeepDockM2	0.772	0.474	0.727	0.455	0.635	0.368
DeepDockM3	0.786	0.478	0.730	0.448	0.635	0.360
RTMScore1	0.807	0.519	0.738	0.485	0.646	0.396
RTMScore2	0.807	0.519	0.745	0.493	0.649	0.409
RTMScore3	0.800	0.506	0.730	0.485	0.653	0.402
Docking (top 20) ^b	0.923	0.667	0.830	0.630	0.775	0.519

^a“_Re” represents that the model is trained based on redocked poses in PDBbind-CrossDocked-Refined while “_Cross” represents that the model is trained based on cross-docked poses. Except for DeepDockM and RTMScore, the data of methods is copied from ref 26.^b The top20 success rate of each docking method for a certain data set, and it represents the upper limit of the value in each column. “re” represents the performance on redocked poses, while “cross” denotes that on cross-docked poses.

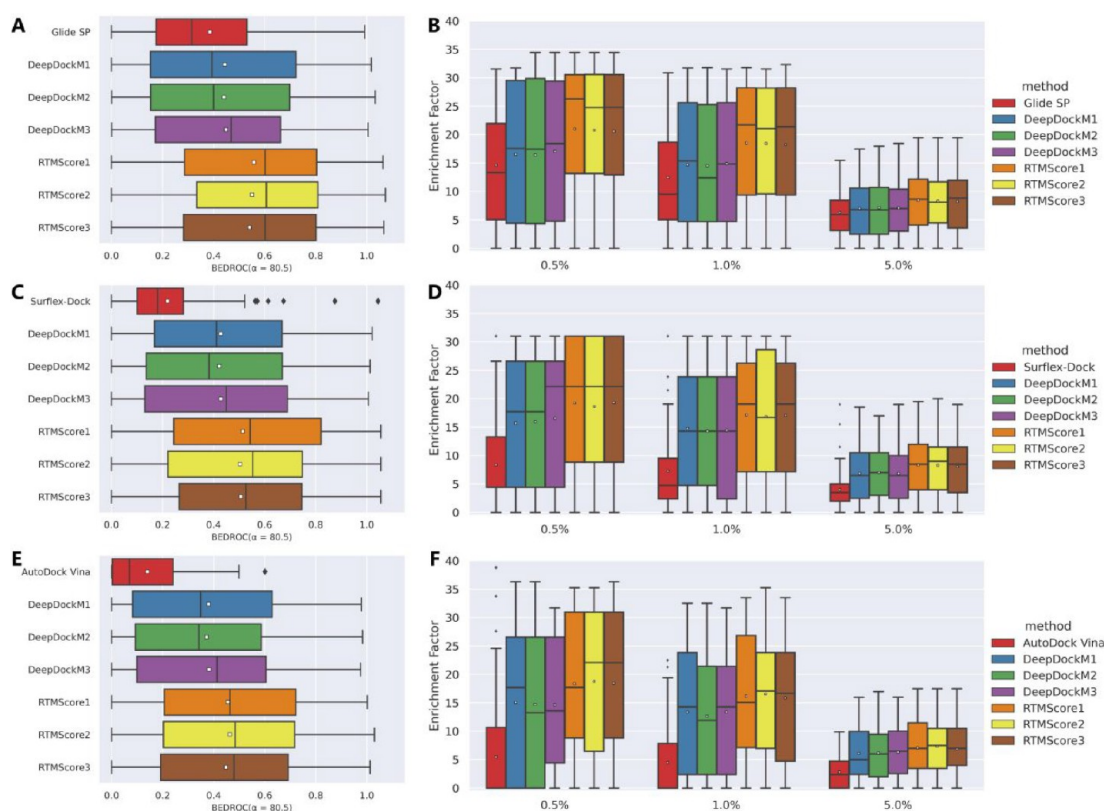


Figure 3. Screening power of scoring functions on the DEKOIS2.0 data set, in terms of (A, C, E) BEDROC ($\alpha = 80.5$) and (B, D, F) enrichment factors at different thresholds (0.5%, 1.0%, and 5.0%), based on the docking poses predicted by (A, B) Glide SP, (C, D) Surflex-Dock, and (E, F) AutoDock Vina. The white square in the box plot denotes the mean value of each statistic.

Table 5. Screening Powers of Scoring Functions on the DEKOIS2.0 Data Set

docking methods	rescoring methods	AUROC		BEDROC ($\alpha = 80.5$)		EF _{0.5%}		EF _{1%}		EF _{5%}	
		mean	median	mean	median	mean	median	mean	median	mean	median
Glide SP		0.747	0.754	0.385	0.314	14.61	13.30	12.47	9.61	6.30	5.97
	DeepDockM1	0.739	0.755	0.444	0.393	16.52	17.57	14.67	15.39	7.02	6.80
	DeepDockM2	0.732	0.753	0.440	0.399	16.41	17.44	14.51	12.40	7.13	6.76
	DeepDockM3	0.741	0.745	0.449	0.468	17.06	18.42	14.90	14.83	7.07	6.99
	RTMScore1	0.768	0.773	0.558	0.602	20.99	26.31	18.53	21.73	8.45	8.64
	RTMScore2	0.768	0.788	0.552	0.606	20.80	24.79	18.43	21.06	8.30	8.12
Surflex-Dock	RTMScore3	0.756	0.762	0.541	0.602	20.56	24.80	18.22	21.36	8.23	8.84
		0.673	0.675	0.220	0.180	8.36	4.43	7.30	4.77	4.00	3.50
	DeepDockM1	0.737	0.759	0.428	0.411	15.69	17.71	14.81	14.31	6.87	6.50
	DeepDockM2	0.727	0.755	0.422	0.382	15.96	17.71	14.28	14.31	7.02	7.00
	DeepDockM3	0.735	0.765	0.428	0.449	16.57	22.14	14.42	14.31	6.83	6.50
	RTMScore1	0.760	0.793	0.514	0.544	19.24	22.14	17.13	19.08	8.33	8.50
AutoDock Vina	RTMScore2	0.764	0.785	0.503	0.554	18.64	22.14	16.84	16.69	8.28	9.00
	RTMScore3	0.752	0.785	0.507	0.527	19.30	22.14	17.07	19.08	8.05	8.50
		0.633	0.637	0.140	0.070	5.46	0.00	4.51	0.00	2.82	2.38
	DeepDockM1	0.707	0.703	0.380	0.349	15.06	17.70	13.36	14.28	6.11	5.00
	DeepDockM2	0.699	0.724	0.372	0.341	14.72	13.29	12.66	11.92	6.18	6.00
	DeepDockM3	0.709	0.724	0.382	0.413	14.63	13.62	13.40	14.28	6.30	6.50
	RTMScore1	0.729	0.760	0.455	0.463	18.36	17.71	16.18	15.08	7.09	7.00
	RTMScore2	0.732	0.752	0.463	0.483	18.76	22.11	16.56	17.11	7.31	7.49
	RTMScore3	0.722	0.763	0.449	0.479	18.47	22.09	15.89	16.69	6.86	6.99

models based on redocked poses on cross-docked poses while the results were substantially the opposite when tested on redocked poses. In addition, it seemed that the models trained on the poses yielded by Surflex-Dock could just perform well on the poses generated by the same docking program, but for the

poses produced by Glide SP and AutoDock Vina, sometimes their performance was even no better than classical SFs. In other words, those ML classifiers are extremely sensitive to the compositions of training sets, whether in terms of the docking mode (redocked or cross-docked) or the docking program used

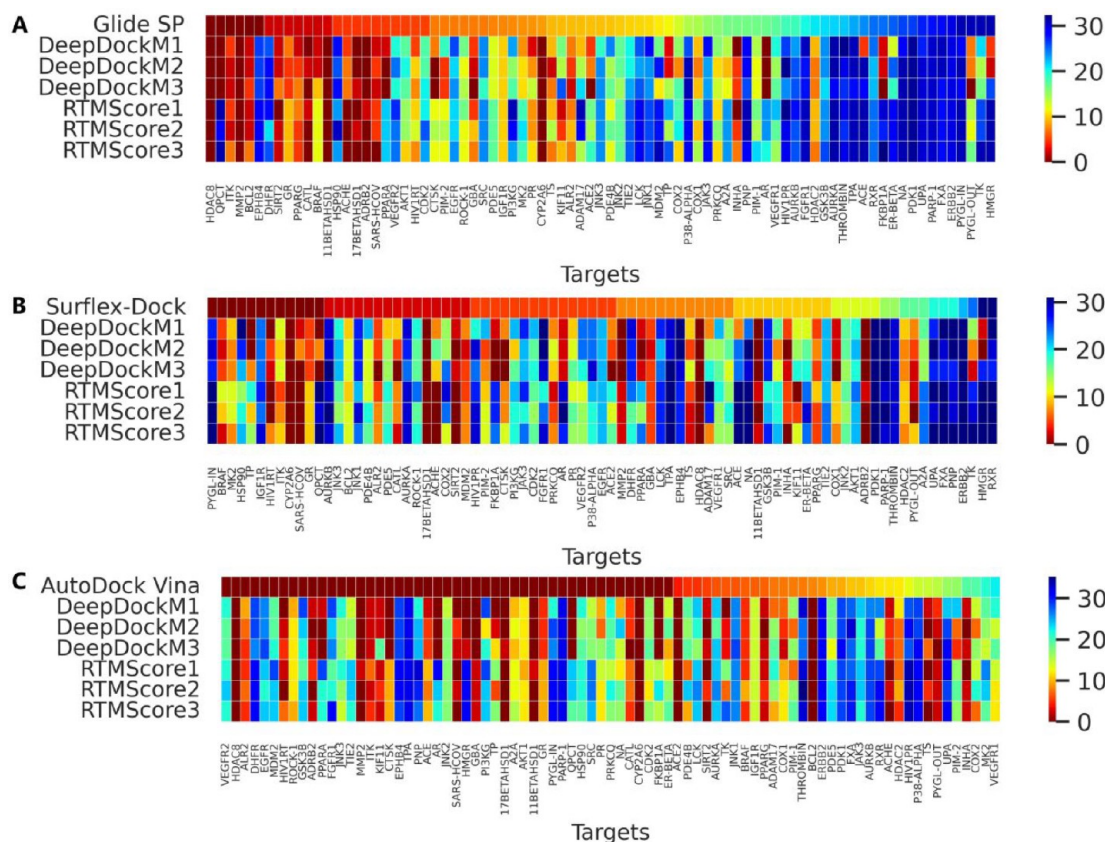


Figure 4. EF_{1%} values of scoring functions on the DEKOIS2.0 data set based on the docking poses predicted by (A) Glide SP, (B) Surflex-Dock, and (C) AutoDock Vina. All of the targets in a subgraph are ranked in descending order in terms of the EF_{1%} values measured by the corresponding docking methods.

for pose generation. However, the method proposed here could obviously achieve a more stable and robust performance. Even if the average performance of RTMScore is compared with the best of these task-specific models, the former just performs slightly worse in terms of the top1 success rate on the cross-docked poses generated by Surflex-Dock (51.8% vs 52.2%) but is still better in the other five cases (80.4% vs 80.0%, 73.8% vs 71.6%, 48.8% vs 47.1%, 64.9% vs 61.4%, and 40.2% vs 36.1%). Furthermore, RTMScore can retain its superiority over DeepDockM, and the improvements on cross-docked poses are more prominent (cross-docked, 51.8% vs 47.7%, 48.8% vs 45.1%, and 40.2% vs 36.7%; redocked, 80.4% vs 78.4%, 73.8% vs 72.7%, and 64.9% vs 64.1%), indicating that RTMScore possesses a better generalization capability on cross-docked poses.

RTMScore Is an Efficient Rescoring Tool for Virtual Screening. Given that the decoys utilized in CASF for the assessment of the screening power are just limited to the crystallized ligands which are unable to bind to a certain target, several larger-scale data sets that involve more active and decoy compounds such as Directory of Useful Decoys (DUD),⁴⁶ Database of Useful Decoys-Enhanced (DUD-E),⁴⁷ and Demanding Evaluation Kits for Objective in Silico Screening (DEKOIS)^{48,49} are usually employed as alternatives for the evaluation of VS methods. Although a cloud of controversy has been thrown upon the reasonability of using these data sets for the assessment of MLSFs due to their hidden biases,^{50,51} considering that our approach is not directly trained on these data sets, the results based on them are still meaningful because the impacts of hidden biases may be equal to all of the tested

approaches. Additionally, the VS performance here can also reflect the generalization capability of our approach to some extent.

Here, a miniature VS data set, i.e., DEKOIS2.0,⁴⁸ was first utilized for the assessment of the screening power of our method, with three popular docking programs (i.e., Glide SP, Surflex-Dock, and AutoDock Vina) serving as the baselines. It should be noted that the ligands failing in docking (especially Glide SP) or rescoring are directly eliminated for the calculation of the final statistics rather than giving an extremely low score. The overall performance in terms of AUROC, BEDROC, and EFs is depicted in Figure S3, Figure 3A,C,E, and Figure 3B,D,F, respectively, and summarized in Table 5, while the statistics across all 81 targets are presented as the heatmaps depicted in Figure 4 and Figures S4–S7. Glide SP could be considered as one of the best-performing docking programs in VS, and it could defeat almost all of the generic MLSFs in our previous assessment on DEKOIS2.0.¹⁷ Here, it could obtain the highest screening power among all three classical methods, with its mean AUROC, BEDROC, EF_{0.5%}, EF_{1%}, and EF_{5%} equaling 0.747, 0.385, 14.60, 12.47, and 6.30, respectively. DeepDockM based on the poses generated by Glide SP could perform no better in terms of the overall indicator AUROC (0.737 ± 0.004), but regarding the other metrics that indicate early recognition capability, it could consistently obtain a superior performance (BEDROC = 0.444 ± 0.004 , EF_{0.5%} = 16.66 ± 0.35 , EF_{1%} = 14.69 ± 0.20 , and EF_{5%} = 7.07 ± 0.06). Compared with DeepDockM, our RTMScore could further gain performance improvements, with the mean AUROC, BEDROC, EF_{0.5%}, EF_{1%}, and EF_{5%} as high as 0.764 ± 0.007 , 0.550 ± 0.009 , 20.78 ± 0.21 , $18.39 \pm$

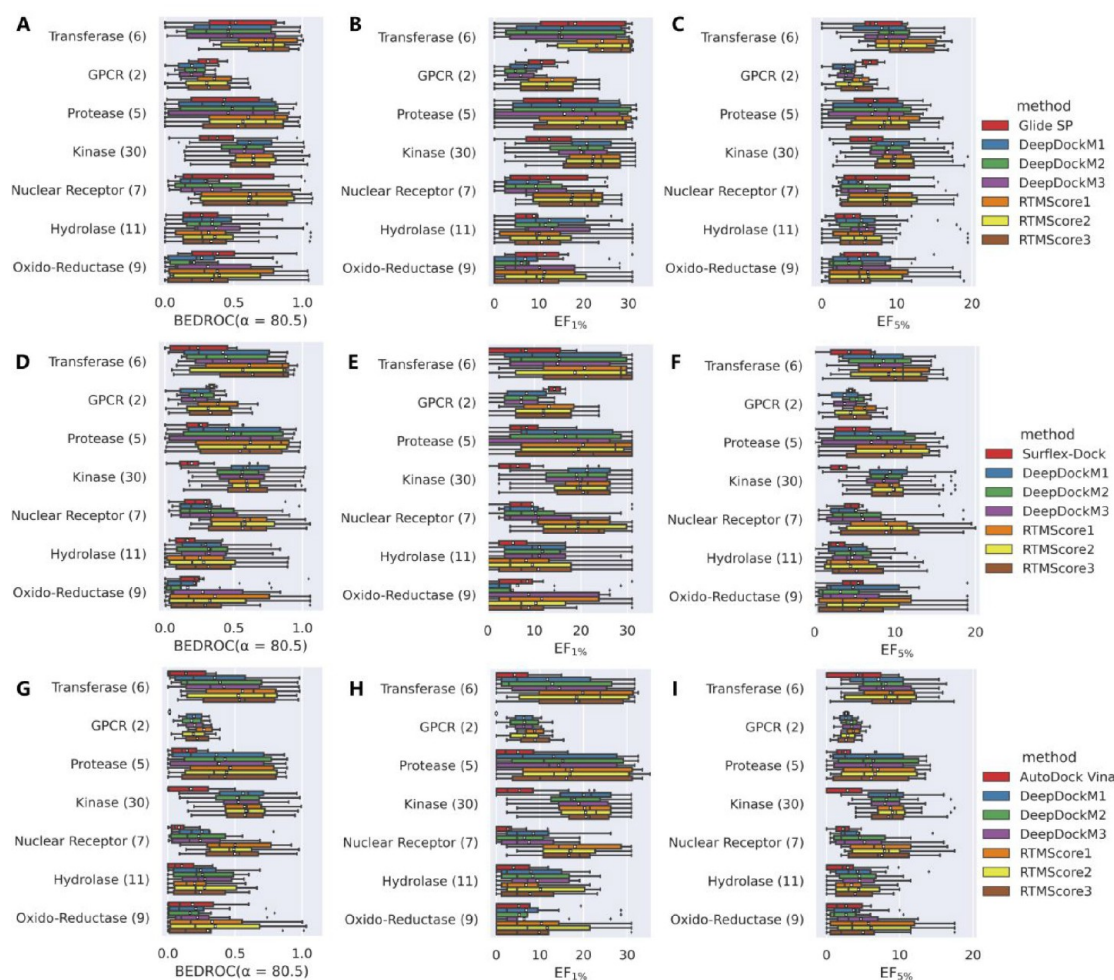


Figure 5. Screening power of scoring functions toward different protein families in the DEKOIS2.0 data set, in terms of (A, D, G) BEDROC ($\alpha = 80.5$), (B, E, H) $EF_{1\%}$, and (C, F, I) $EF_{5\%}$, based on the docking poses predicted by (A–C) Glide SP, (D–F) Surflex-Dock, and (G–I) AutoDock Vina. The white square in the box plot denotes the mean value of each statistic. Isomerases, ligase, and other are omitted because they only contain one target.

0.16, and 8.33 ± 0.11 , respectively. When the poses generated by Surflex-Dock or AutoDock Vina were used for rescoring, despite a certain degree of performance decrease, RTMScore could still achieve improved performance compared with the corresponding docking methods and could even outperform Glide SP and DeepDockM in terms of all of the metrics. The comparison based on the median statistics showed a substantially similar trend, so that will not be covered again here.

We also investigated the performance (BEDROC, $EF_{1\%}$, and $EF_{5\%}$) of each method on the individual target families, as reported in Figure 5. The 81 targets in the DEKOIS2.0 data set can be split into 10 families, and the isomerase, ligase, and other containing just one target are omitted. As we can see, RTMScore can consistently outperform classical methods on five families (i.e., transferase, protease, kinase, nuclear receptor, and hydrolase) but is slightly worse than (or comparable with) Glide SP on the other two families. The GPCR family consists of only two targets; it is expected to see the fluctuant performance on this family. As for oxido-reductase, we further examined the structures and found that, for several targets (e.g., INHA and 11BSTAHS1), a substrate other than the cocrystallized ligand also occupied some part of the pocket and might form interactions with the ligand. The existence of the substrate may influence the rescoring of our approach, thus leading to its worse performance for some targets.

To further verify the efficiency of our approach in VS, we also turn to a larger data set, i.e., DUD-E, for evaluation. DUD-E is composed of 102 targets from 8 diverse protein families, containing a total of 22 886 active compounds with around 50 decoys for each. Here, we employed the previously best-performing docking program, i.e., Glide SP, to generate the poses first and recorded the rescoring results in Figure 6 and Table 6. Whether in terms of mean or median statistics, here, DeepDockM (mean BEDROC = 0.422 ± 0.010 , $EF_{0.5\%} = 30.94 \pm 0.87$, $EF_{1\%} = 25.03 \pm 0.85$, and $EF_{5\%} = 8.94 \pm 0.15$) performs no better than Glide SP (mean BEDROC = 0.414, $EF_{0.5\%} = 29.44$, $EF_{1\%} = 23.61$, and $EF_{5\%} = 9.24$), but our approach still exhibits its superiority (mean BEDROC = 0.548 ± 0.013 , $EF_{0.5\%} = 41.73 \pm 1.03$, $EF_{1\%} = 34.19 \pm 0.95$, and $EF_{5\%} = 10.81 \pm 0.19$). The performance in terms of individual target family is depict in Figure 7, where we can still observe the universally superior performance of our approach over Glide SP and DeepDockM.

Of course, some models trained with the involvement of a lot of decoy compounds and/or trained in a target-specific way may exhibit more striking results on the DUD-E or DEKOIS2.0 data set. For example, InteractionGraphNet⁵² trained on the DUD-E data set could obtain the BEDROC of 0.762, $EF_{0.5\%}$ of 22.46, and $EF_{1\%}$ of 22.23 on the DEKOIS2.0 data set. SIEVE-Score⁵³ trained in a target-specific way could obtain an average $EF_{1\%}$ of 43.913 on the DUD-E data set. The models proposed by Ragoza

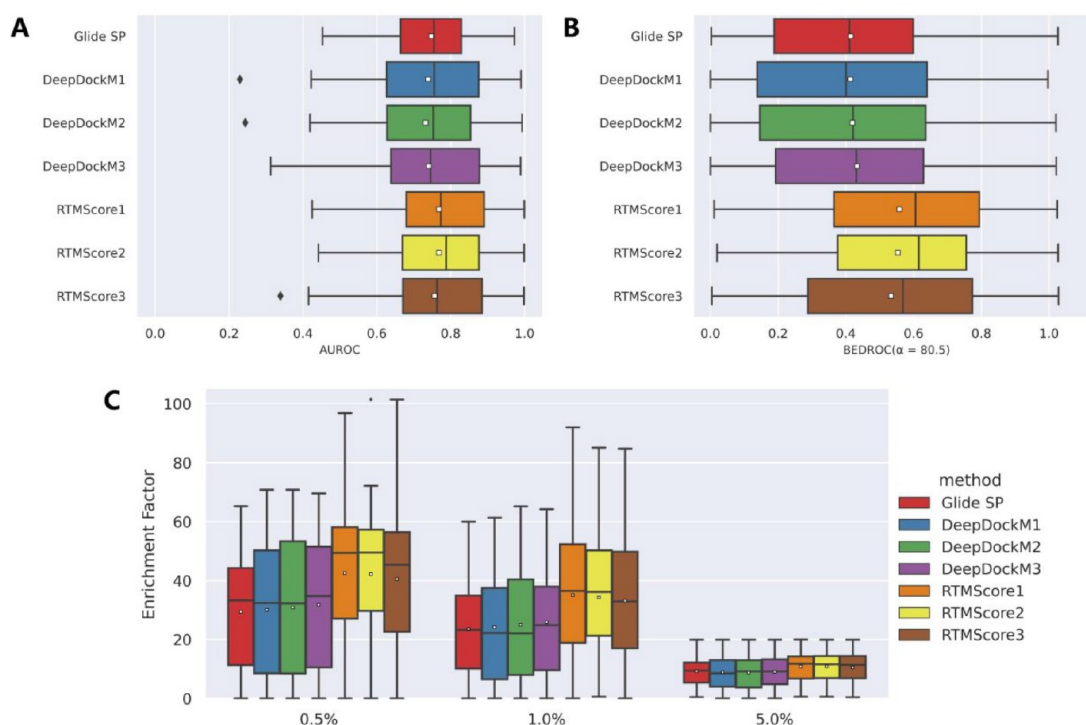


Figure 6. Screening power of scoring functions on the DUD-E data set, in terms of (A) AUROC, (B) BEDROC ($\alpha = 80.5$), and (C) enrichment factors at different thresholds (0.5%, 1.0%, and 5.0%), based on the docking poses predicted by Glide SP. The white square in the box plot denotes the mean value of each statistic.

Table 6. Screening Powers of Scoring Functions on the DUD-E Data Set

docking methods	rescoring methods	AUROC ($\alpha = 80.5$)		BEDROC		EF _{0.5%}		EF _{1%}		EF _{5%}	
		mean	median	mean	median	mean	median	mean	median	mean	median
Glide SP		0.820	0.845	0.414	0.410	29.44	33.27	23.61	23.30	9.24	9.50
	DeepDockM1	0.796	0.819	0.413	0.400	30.09	32.43	24.21	22.22	8.91	8.56
	DeepDockM2	0.783	0.817	0.419	0.421	30.90	32.36	24.96	22.14	8.80	9.24
	DeepDockM3	0.798	0.830	0.432	0.431	31.84	34.77	25.91	24.86	9.10	9.27
	RTMScore1	0.830	0.863	0.558	0.606	42.47	49.42	35.10	36.54	10.88	11.82
	RTMScore2	0.831	0.869	0.553	0.615	42.17	49.53	34.28	36.18	10.95	11.65
	RTMScore3	0.825	0.862	0.533	0.568	40.56	45.37	33.20	32.95	10.59	11.44

et al.,⁵⁴ Torng et al.,⁵⁵ and Lim et al.⁵⁶ could obtain ROC enrichment scores at a 1.0% threshold of 29.654, 29.748, and 69.037, respectively, when a cross-validation is conducted on DUD-E. However, their performance may be significantly overestimated, since their training ways make the model easily memorize the hidden biases in the training sets rather than generalization. Some studies have reported their poor generalization on the test sets that possess completely different compositions (e.g., different targets, different ratios of actives to decoys, and different approaches to generate the decoys) as the training set.^{17,18} In contrast, the training set of our approach just comprises experimentally determined protein–ligand complexes, and the excellent performance here can in turn confirm the robustness of the method.

To summarize, all of the above results demonstrate that RTMScore is indeed an efficient rescoring tool to improve the screening power of conventional docking programs in VS. However, it should be noted that this improvement may not be applicable to all targets, just as shown in Figure 4 and Figures S4–S8. One reason is that this knowledge-like SF has a substantially different principle from the empirical SFs utilized in

Glide SP, Surflex-Dock, or AutoDock Vina. Empirical SFs identify actives from decoys according to their binding affinities, while our method exhibits poor scoring and ranking powers, and it just relies on the reasonability of protein–ligand binding conformations to make the judgment. Hence, if a specific target is focused on, perhaps an extra assessment is still necessary.

RTMScore Is Insensitive to the Rotation of Protein. We also conducted a simple experiment to explore whether our approach is sensitive to the rotation of proteins. Two complexes (PDB entries: 1uto and 4f9w) were randomly selected from the CASF test set and rotated by every 30° around the X-axis, Y-axis, and Z-axis, respectively, centered on the origin. The rotation of the structures was executed using an in-house script based on the Python API available in Schrödinger 2022. As shown in Figure 8, we can observe a certain degree of the fluctuation of the scores predicted by DeepDockM while the scores of RTMScore remain unchanged. These findings are expected, since the protein graphs in DeepDockM utilize relative Cartesian coordinates as edges which are susceptible to the coordinate transformations, while the initial spatial representations for our method are just limited to rotation-insusceptible distances and dihedral angles.

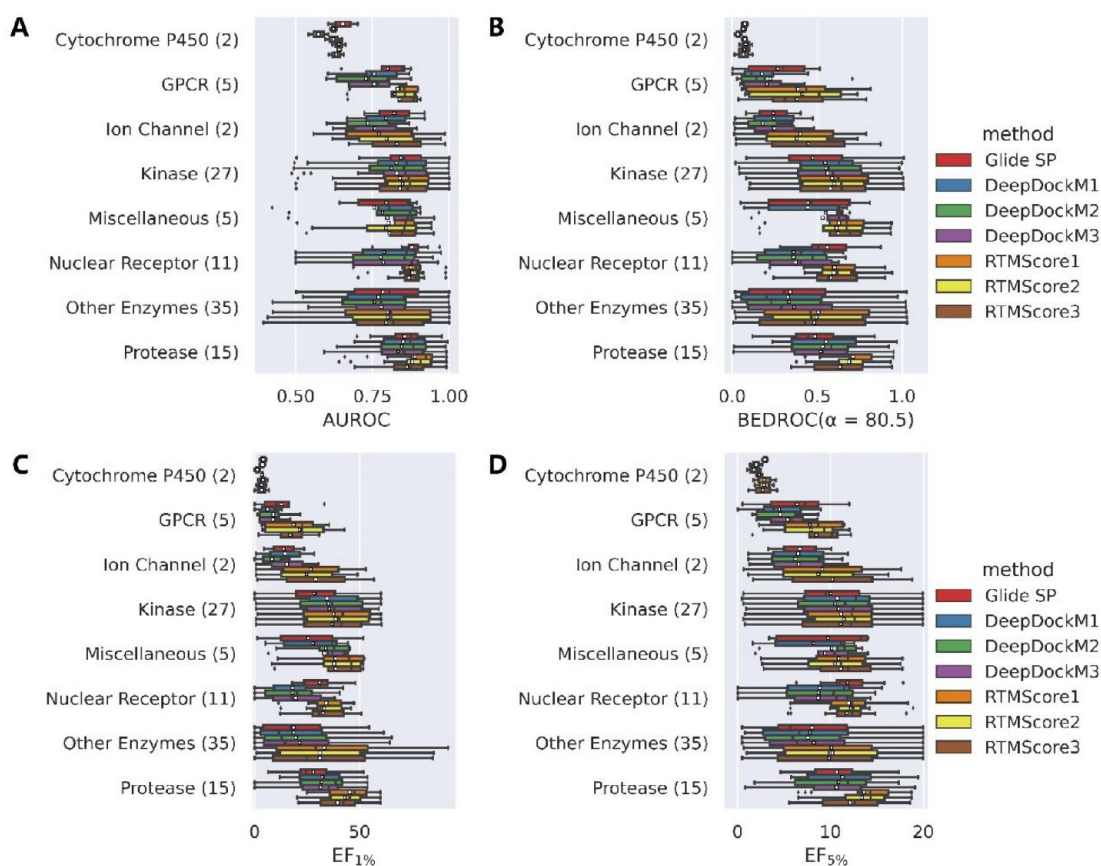


Figure 7. Screening power of scoring functions toward different protein families in the DUD-E data set, in terms of (A) AUROC, (B) BEDROC ($\alpha = 80.5$), (C) $EF_{1\%}$, and (D) $EF_{5\%}$, based on the docking poses predicted by Glide SP. The white square in box plot denotes the mean value of each statistic. Isomerases, ligase, and other are omitted because they only contain one target.

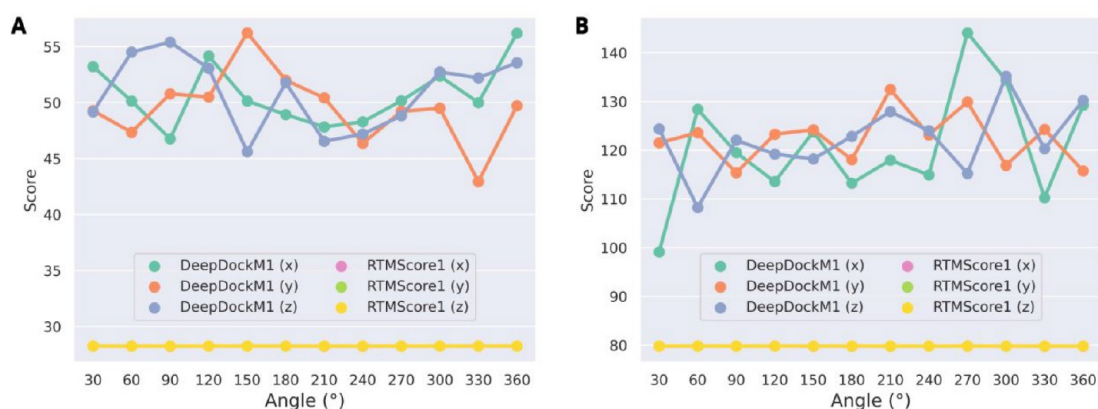


Figure 8. Fluctuation of the predicted scores of the (A) 1uto and (B) 4f9w complexes. *x*, *y*, and *z* represent that the complexes are rotated by every 30° around the X-axis, Y-axis, and Z-axis, respectively, centered on the origin.

The stability of the scores suggests that the excellent performance of our approach could be retained as the coordinates of the proteins transform, which can be considered as another superiority over DeepDock.

RTMScore Can Provide Extra Information at the Atom or Residue Level. Since RTMScore adopts an additive function form to integrate all independent residue–atom distance likelihood values into the final statistical potential, it can be easily decomposed into the contribution from each atom in a ligand or each residue in a protein pocket. Although the score of our approach could hardly reflect the real binding

affinity of a protein–ligand complex, it can surely reflect the stability of the formed 3D binding conformation. Here, we take androgen receptor (AR) as an example to illustrate how these scores at the atom or residue level can express. Figure 9A depicts five ligands of AR retrieved from the CASF benchmark, and the atom with darker color and denser contour contributes more to the total score of the ligand. As can be seen, the atoms consisting of the 4-cyano-3-trifluoromethyl (or halogeno) phenyl fragment have relatively higher scores, suggesting that this region may play a more important role in their binding with the target. In fact, this fragment has been widely adopted in approved AR

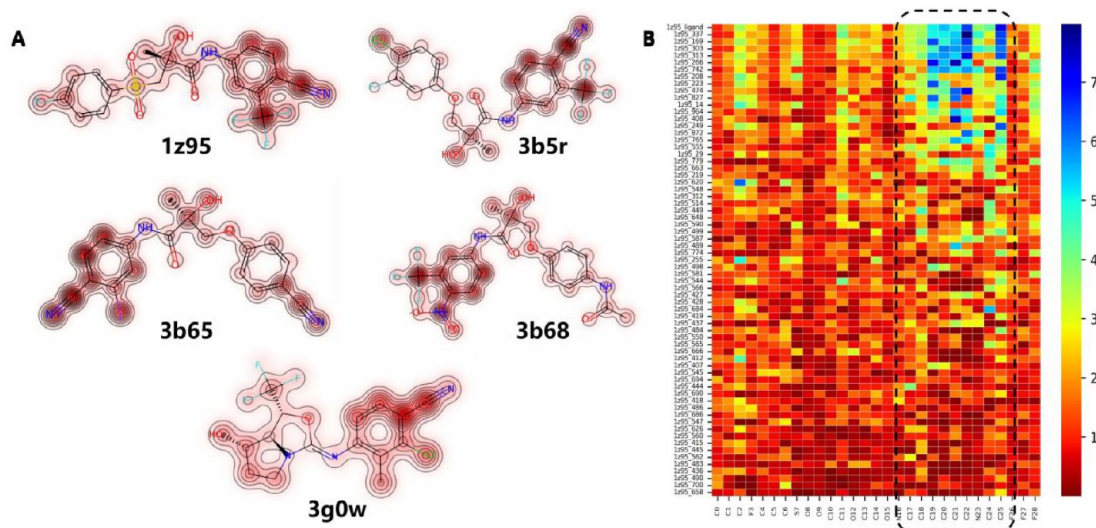


Figure 9. RTMScore decomposed into contributions at the atom level. (A) Distributions of the atomic contributions for the five cocrystallized ligands of androgen receptor (AR) existing in the CASF-2016 test set. The darker color represents the higher score. (B) Atomic contributions of the multiple docking poses for a representative ligand (PDB entry: 1z95) of AR. Red and blue represent the lower and higher scores, respectively. The X axis represents each atom in the ligand of 1z95, and the Y axis denotes each pose. The poses are ranked according to their RMSD values relative to native poses in ascending order.

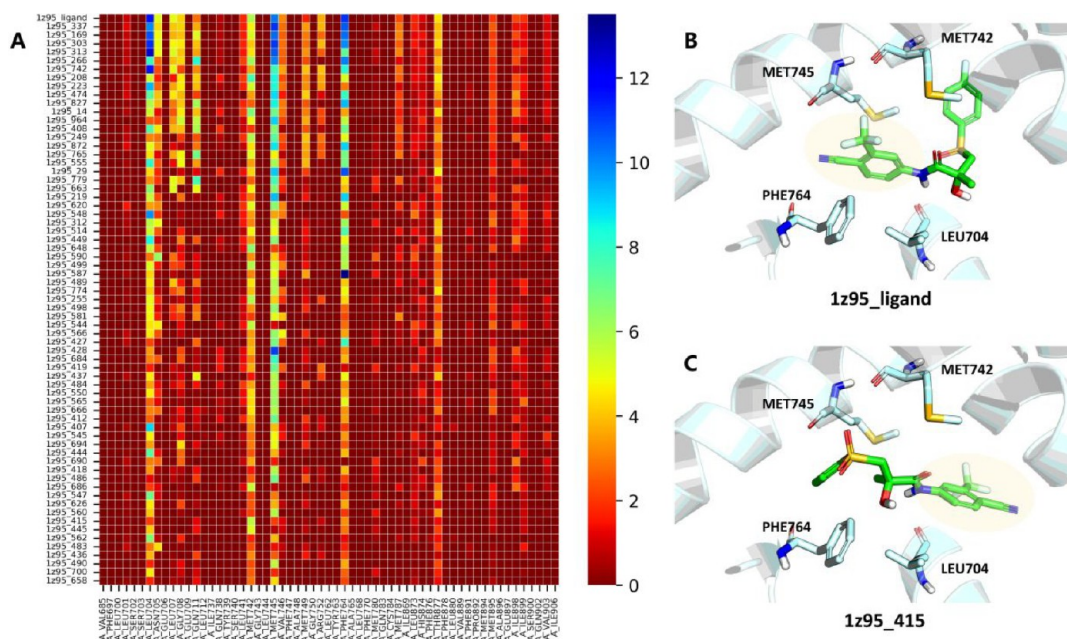


Figure 10. RTMScore decomposed into contributions at the residue level. (A) Residue contributions of the multiple docking poses for a representative ligand (PDB entry: 1z95) of AR. Red and blue represent the lower and higher scores, respectively. The X axis represents each residue in the protein pocket, and the Y axis denotes each pose. The poses are ranked according to their RMSD values relative to native poses in ascending order. (B) Native binding pose of the cocrystallized ligand of 1z95 (1z95_ligand). (C) Representative incorrect binding of 1z95 (1z95_415).

antagonists such as bicalutamide and enzalutamide and is recognized as the core pharmacophore for this series of compounds.⁵⁷ Figure 9B shows the atomic contributions of each docking pose of 1z95 in the CASF docking set, and one can observe that the poses with higher scores of the 4-cyano-3-trifluoromethyl phenyl region are always closer to the native pose, further verifying the importance of this region on the formation of reasonable binding poses.

We can also interpret the model from the perspective of residue, as shown in Figure 10A. Similarly, the residues contributing more to the binding of the ligand, including

LEU704, MET742, MET745, and MET764, are given higher weights, and the poses with the higher scores of these residues tend to form a more reasonable binding conformation judging by the comparison with the native pose. The native pose of 1z95 is depicted in Figure 10B, and we can find that these four residues exactly encompass the 4-cyano-3-trifluoromethyl phenyl region of the ligand, thus partially explaining why these residues seem so important. Figure 10C is a representative incorrect pose, whose 4-cyano-3-trifluoromethyl phenyl region is far from the four residues, correlating to its low score.

CONCLUSIONS

In this work, we have developed a residue–atom distance likelihood-based SF named RTMScore for protein–ligand binding pose prediction. Our approach inherits the overall architecture of DeepDock, which consists of a feature extraction module to obtain the node representations of proteins and ligands, a feature concatenation module to concatenate all node features, and an MDN to learn the probability distribution of the distance for each ligand–protein pair, but introduced a tailored residue-based graph representation strategy for the encoding of protein pockets and several graph transformer layers for the further feature extraction of both the proteins and ligands. These advances facilitate a remarkable improvement of our method in both the binding pose prediction and VS tasks in terms of both the prediction accuracy and generalization capability. Finally, in addition to the conventional interpretation from the perspective of atomic contributions, the residue-based graph representation enables our method to decompose the scores at the residue level, demonstrating its better interpretability.

Considering its robust performance and wider application domain, we can expect the promising prospect of RTMScore in structure-based drug design. On one hand, it not only can directly combine with popular docking programs and serve as a rescoring tool for either binding pose prediction or VS but also can be embedded into existing docking tools (e.g., AutoDock Vina) just like Gnina⁵⁸ to become a new docking tool. On the other hand, it can be coupled with mainstream molecular generation algorithms to develop new protocols for *de novo* drug design, which can be considered in future work.

EXPERIMENTAL SECTION

Data Set Preparation. The latest version of the PDBbind database (v2020)⁵⁹ that contains a total of 19 443 protein–ligand complexes in its general set was utilized here for model construction. Considering that there are quite a few errors in the structures directly retrieved from PDBbind, we reprocessed the structures with the Protein Preparation Wizard⁶⁰ module in Schrödinger 2020, including removing waters, adding hydrogens, assigning bond orders, optimizing H-bond networks, and minimizing the systems using the OPLS3⁶¹ force field until the RMSD of heavy atoms averaged at 0.30 Å. Meanwhile, we carefully checked the prepared structures to guarantee that they could be recognized by the RDKit¹⁹ toolkit as much as possible. Then, we excluded the entries intersecting with those from the PDBbind-v2020 core set and CASF-2016 benchmark, thus resulting in a total of 19 149 complexes.

Graph Representations for Ligands and Proteins. Similar with DeepDock, the 2D structures of the ligands were described as undirected graphs [$G_l = (V_l, E_l)$], where the nodes and edges represented the atoms and bonds in a molecule, respectively. Table 1 summarizes the input node and edge features for the ligands, which have been widely employed in previous studies to represent small molecules.^{52,62}

To save computational resources, the proteins were first converted to the form of binding pockets that were defined as the residues within 10.0 Å around the cocrystallized ligands, where the cutoff of 10.0 Å is empirical and optimizable. Then, the pockets can be represented as undirected graphs [$G_p = (V_p, E_p)$], where the nodes represent the residues in a pocket while the edges represent the interaction between any two residues whose minimum distances are less than 10.0 Å. We did not employ a treatment similar to that for the ligands (i.e., atom and bond) to define the pockets, since it was obvious that noncovalent interactions played a more important role than covalent bonds to maintain the 3D structures of the proteins.^{63,64} Considering that the calculations of the interactions for all atomic pairs might be too resource-intensive, here, we just simplified each protein pocket as a residue-based graph. The input node and edge features for protein

graphs could be found in Table 2. In addition to the 20 standard amino acids, we also included 10 nonstandard amino acids existing in the proteins from PDBbind-v2020 for encoding. To better describe the geometry configuration of each residue, we further incorporated five distance-based features (including the maximum and minimum distances within any two atoms in a residue, the distances between the CA and O atoms, the O and N atoms, and the C and N atoms) and four dihedral angles (including *phi*, *psi*, *omega*, and *chi1*). The schematic view of the representation of each residue is depicted in Figure 1A. Regarding edge features, we employed four types of distances, i.e., the distance between the CA atoms of two residues, the distance between the centers of mass of two residues, and the maximum and minimum atomic distances between two residues. The above geometric features are all invariant to rotation and translation of the protein–ligand complex, thus making the model insensitive to the input complexes with different spatial coordinates. The features were calculated majorly based on RDKit¹⁹ and MDAnalysis,⁶⁵ and the graphs were generated through the Deep Graph Library (DGL)⁶⁶ package.

Model Architecture. The overall model architecture of RTMScore is shown in Figure 1C, which is composed of a feature extraction module, a feature concatenation module, and an MDN.⁶⁷

Feature Extraction Module. Here, we employ an expanded graph transformer framework, which is similar to the architecture proposed by Dwivedi et al.⁶⁸ and Morehead et al.,⁶⁹ to learn the node representations for G_l and G_p . It should be noted that the modules for the protein and ligand are independent, although they share the same architecture. Before being fed into the tandem repeated graph transformer layers, the input node features $\alpha_i \in \mathbb{R}^{d_h \times 1}$ for node i and edge features $\beta_{ij} \in \mathbb{R}^{d_e \times 1}$ for the edge between node i and its neighboring node j should be first embedded into d -dimensional initial hidden representations h_i^0 and e_{ij}^0 through two independent linear projections:

$$h_i^0 = W_\alpha^0 \alpha_i + b_\alpha^0, \quad e_{ij}^0 = W_\beta^0 \beta_{ij} + b_\beta^0 \quad (1)$$

where $W_\alpha^0 \in \mathbb{R}^{d \times d_h}$, $W_\beta^0 \in \mathbb{R}^{d \times d_e}$, and $b_\alpha^0, b_\beta^0 \in \mathbb{R}^d$ are the weights and biases of linear projection layers. The overall architecture of the graph transformer layer utilized here is depicted in Figure 1B, which mainly relies on a modified multihead self-attention (MHA) mechanism to update the node and edge features of each graph. The details for the update operation of the l th layer are described as the following equations:

$$q_{ij}^{k,l} = Q^{k,l} \text{Norm}(h_i^l) \quad (2)$$

$$k_{ij}^{k,l} = K^{k,l} \text{Norm}(h_j^l) \quad (3)$$

$$v_{ij}^{k,l} = V^{k,l} \text{Norm}(h_j^l) \quad (4)$$

$$e_{ij}^{k,l} = E^{k,l} \text{Norm}(e_{ij}^l) \quad (5)$$

$$\omega_{ij}^{k,l} = \text{Softmax}_j \left(\left(\frac{q_{ij}^{k,l} \cdot k_{ij}^{k,l}}{\sqrt{d_k}} \right) \cdot e_{ij}^{k,l} \right) \quad (6)$$

$$\hat{h}_i^{l+1} = h_i^l + O_{h1}^l \text{Dropout} \left(\text{Concat}_{k \in 1, \dots, H} \left(\sum_{j \in N_i} \omega_{ij}^{k,l} v_{ij}^{k,l} \right) \right) \quad (7)$$

$$\hat{e}_{ij}^{l+1} = e_{ij}^l + O_{e1}^l \text{Dropout}(\text{Concat}_{k \in 1, \dots, H}(\omega_{ij}^{k,l})) \quad (8)$$

$$h_i^{l+1} = \hat{h}_i^{l+1} + O_{h2}^l \text{Dropout}(\text{SiLU}(O_{h1}^l \text{Norm}(\hat{h}_i^{l+1}))) \quad (9)$$

$$e_{ij}^{l+1} = \hat{e}_{ij}^{l+1} + O_{e2}^l \text{Dropout}(\text{SiLU}(O_{e1}^l \text{Norm}(\hat{e}_{ij}^{l+1}))) \quad (10)$$

where $Q^{k,l}, K^{k,l}, V^{k,l}, E^{k,l} \in \mathbb{R}^{d_k \times d}$, $O_{h1}^l, O_{e1}^l \in \mathbb{R}^{d \times d}$, $O_{h2}^l, O_{e2}^l \in \mathbb{R}^{d \times 2d}$, and $\phi_{h1}^l, \phi_{e1}^l \in \mathbb{R}^{2d \times d}$, and $\phi_{h2}^l, \phi_{e2}^l \in \mathbb{R}^{d \times 2d}$ are all learnable parameters for linear layers; $k \in 1, \dots, H$ denotes the number of attention heads; d_k represents the dimension

of each head, which equals d divided by H ; $j \in N_i$ represents the neighboring nodes of node i ; Norm denotes batch normalization; Concat denotes the concatenation operation; Dropout denotes the dropout operation; SiLU is a type of nonlinear activation; and Softmax denotes the softmax operation. For numerical stability, the softmax outputs were clamped to a value between -5 and $+5$. Specifically, the hidden representations h_i^l and e_j^l were first normalized by a batch normalization layer, and then, the corresponding Query ($q_{ij}^{k,l}$), Key ($k_{ij}^{k,l}$), and Value ($v_{ij}^{k,l}$) in MHA, as well as the hidden representation for edge ($e_{ij}^{k,l}$), could be calculated through eqs 2–5. Subsequently the attention score $\omega_{ij}^{k,l}$ could be obtained through eq 6, followed by the calculation of $\hat{h}_i^{l+1}$ and $\hat{e}_j^{l+1}$, which represent the intermediate node and edge representations, respectively. Finally, $\hat{h}_i^{l+1}$ and $\hat{e}_j^{l+1}$ should go through several batch normalization layers, feed forward networks (FFNs), and residual connections, thus leading to the output node and edge representations of the l th layer (or the input representations of the $(l+1)$ th layer) h_i^{l+1} and e_j^{l+1} .

Feature Concatenation Module and Mixture Density Network. These two modules are substantially the same as those in DeepDock. The learned node representations for protein h_p^{prot} and ligand h_s^{lig} (where p and s represent the node index of the protein and ligand, respectively) were pairwise-concatenated to obtain the input for MDN. The major operations in MDN can be described as follows:

$$h_{p,s} = \text{Dropout}(\text{ELU}(\text{Norm}(W_c \text{Concat}([h_p^{\text{prot}}, h_s^{\text{lig}}]))) \quad (11)$$

$$\mu_{p,s} = \text{ELU}(W_\mu h_{p,s}) + 1 \quad (12)$$

$$\sigma_{p,s} = \text{ELU}(W_\sigma h_{p,s}) + 1.1 \quad (13)$$

$$\rho_{p,s} = \text{Softmax}(W_\rho h_{p,s}) \quad (14)$$

where $W_c \in \mathbb{R}^{2d \times d_m}$, W_μ , W_σ , and $W_\rho \in \mathbb{R}^{d_m \times N_g}$ are learnable weights for linear layers, and here, the bias terms are omitted for clarity of presentation. Concat, Softmax, and Dropout denote the concatenation, softmax, and dropout operations, respectively. The concatenated features of proteins and ligands were first converted to d_m -dimensional hidden representations in MDN ($h_{p,s}$) through an FFN, which consists of a linear layer, a batch normalization, and an exponential linear unit (ELU)⁷⁰ as the activation function. Then, three parameter vectors to determine a mixture density model, including means ($\mu_{p,s}$), standard deviations ($\sigma_{p,s}$), and mixing coefficients ($\rho_{p,s}$) could be computed through eqs 12–14. This mixture density model can be considered as the mixture of N_g Gaussians, thus mimicking the probability density distribution of the distance between each ligand node and protein node. Considering that protein was represented at the residue level, here, we tried three types of distances, i.e., the minimum distance between a specific ligand atom and each residue atom, the distance between a specific ligand atom and the CA atom of a residue, and the distance between a specific ligand atom and the center of mass of a residue, and finally, the first one was selected as the indicator.

Inspired by the study reported by Méndez-Lucio et al.,³⁶ we also introduced two auxiliary tasks named atom type and bond type, which were calculated based on the previously learned ligand node representation h_s^{lig} . These two tasks were used to learn the atom type (C, N, O, S, F, P, Cl, Br, I, B, Si, Fe, Zn, Cu, Mn, Mo, or other) of each atom and bond type (SINGLE, DOUBLE, TRIPLE, or AROMATIC) of each bond connecting the specific node and its neighboring nodes, thus forcing the model to learn molecular structures.

Model Training. The prepared PDBbind-v2020 was randomly split into a validation set composed of 1500 complexes and a training set with 17 649 complexes. Considering the randomness of random split as well as the model training, we trained three independent models with different random seeds (126, 542, and 23). The models were optimized using the Adam optimizer, with a batch size of 128, a learning rate of 10^{-3} , and a weight decay of 10^{-5} . The model training proceeded unless the validation loss did not change in 70 successive epochs. The detailed hyperparameters for the models trained here can be found in Table S1.

The loss function is defined as eq 16, which is composed of an MDN loss ($\mathcal{L}_{\text{MDN}}$) and two cross-entropy losses to reflect the capabilities of

the model to predict atom and bond types ($\mathcal{L}_{\text{at}}$ and $\mathcal{L}_{\text{bt}}$), respectively. A weight of 0.001 was given to these two auxiliary tasks to balance the losses. As shown in eq 17, $\mathcal{L}_{\text{MDN}}$ is defined as the negative log-likelihood of $d_{p,s}$ that denotes the distance between protein node h_p^{prot} and ligand node h_s^{lig} , implying that minimizing $\mathcal{L}_{\text{MDN}}$ is to find the most likely distance that could separate the h_p^{prot} from h_s^{lig} . The probability was calculated through the mixture density model N formed by N_g Gaussians and parametrized by $\rho_{p,s}$, $\mu_{p,s}$, and $\sigma_{p,s}$. Then, the statistical potential for a specific protein–ligand complex could be calculated by summing up the negative log-likelihood values of all potential protein–ligand node pairs (eq 18). It should be noted that only the protein–ligand node pairs with their $d_{p,s} \leq 7.0$ Å were considered when training the models. The cutoff of 7.0 Å was also an optimizable parameter, and here, we just retained the same setting as DeepDock. Despite this, when using the trained models to score the protein–ligand complexes, Méndez-Lucio et al. found that the scores without the contributions of the node pairs whose distances were larger than 3.0 Å could yield the optimal performance. We adopted a similar strategy to tune the cutoff and finally neglected the node pairs with their distances larger than 5.0 Å when using the models for predictions.

$$\mathcal{L} = \mathcal{L}_{\text{MDN}} + 0.001 \times \mathcal{L}_{\text{at}} + 0.001 \times \mathcal{L}_{\text{bt}} \quad (16)$$

$$\begin{aligned} \mathcal{L}_{\text{MDN}} &= -\log P(d_{p,s} | h_p^{\text{prot}}, h_s^{\text{lig}}) \\ &= -\log \sum_{n=1}^{N_g} \rho_{p,s,n} N(d_{p,s} | \mu_{p,s,n}, \sigma_{p,s,n}) \end{aligned} \quad (17)$$

$$U_{(x)} = -\sum_{p=1}^P \sum_{s=1}^S \log P(d_{p,s} | h_p^{\text{prot}}, h_s^{\text{lig}}) = -\text{score} \quad (18)$$

Model Evaluation. The CASF benchmark originally proposed by Cheng et al.⁷¹ was considered as the gold standard for the assessment of classical SFs. The latest CASF-2016³⁷ that contains 285 diverse protein–ligand complexes was employed here for the initial evaluation of our method. There are four different tasks assessed in CASF-2016, including the scoring, docking, ranking, and screening powers. The detailed description of the methods/protocols can be found in its original publication. Considering that only the protein–ligand complexes are involved in the training of distance likelihood-based methods, the scoring and ranking powers that are highly relevant with the binding affinity data are not so applicable. In fact, unless incorporated as an empirical term to fit the experimental binding affinities, most knowledge-based SFs perform rather poorly in terms of the scoring and ranking powers.^{41,72} Hence, we mainly focused on the docking and screening powers in this study. The docking task was evaluated primarily according to success rate (SR), a metric where the complex can be marked as a successful prediction if one of the RMSD values between the best-scored poses and the native pose is below the predefined threshold (usually 2.0 Å). The indicator of the screening power includes the SR of identifying the highest-affinity binder among the 1%, 5%, or 10% top-ranked ligands whether in “forward screening” or “reverse screening”, and the enrichment factor (EF) which is defined as the percentage of true binders observed among all of the true binders for a given percentile of the top-ranked candidates (1%, 5%, or 10%) of a chemical library.

In addition to CASF, several extra data sets were also selected to further estimate the generalization capability of the method in terms of the docking and screening powers. Among them, DEKOIS2.0⁴⁸ and DUD-E⁴⁷ were utilized for the further investigation on the screening power. The former consists of 81 structurally diverse targets, each containing 40 active ligands and 1200 decoys, while the latter is made up of 102 targets, containing a total of 22 886 active compounds with around 50 decoys for each. The proteins were reprocessed with the Protein Preparation Wizard module as described above, and the prepared ligands are directly employed for docking. Three popular docking programs, namely, AutoDock Vina (version 1.1.2),³⁸ Glide SP (version 86012),⁴⁰ and Surflex-Dock (version 2.706.13302),⁴² were used first to generate the potential binding poses for the compounds in

DEKOIS2.0 while, for the molecules in DUD-E, only Glide SP was employed, and up to 10 poses remained for each ligand. The detailed settings for each docking program can be found in ref 73. Then, these poses were rescored by RTMScore, and the best RTMScore score for the 10 poses was employed as the final score for this compound. The screening power was estimated majorly in terms of area under the receiver operating characteristic curve (AUROC),⁷⁴ Boltzmann enhanced discrimination of receiver operating characteristic (BEDROC, $\alpha = 80.5$),⁷⁵ and EFs at different thresholds (0.5%, 1%, and 5%). As for the docking power, we also tested our method on several cross-docking data sets constructed in our previous study.²⁶ Compared with redocked poses that are widely utilized in the assessment of docking power, cross-docked poses are closer to the real-world scenarios, thus reflecting the generalization capability to some extent.

For the sake of comparison, we also retrained DeepDock (denoted as DeepDockM) on our data set. We only changed the graph representation for ligands to our own mode while keeping other parameters/architectures unchanged. In addition, we also trained several models by combining our residue-based graph representation for proteins and model architecture adopted in DeepDock (denoted as RTMScoreM), in order to explore if graph transformer consistently works for the extraction of node representations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.2c00991>.

Hyperparameter setting of each model; scoring and ranking powers of scoring functions on the CASF-2016 benchmark; docking power of scoring functions on the CASF-2016 benchmark in terms of binding funnel analysis; AUROC values of scoring functions on the DEKOIS2.0 data set; AUROC values of scoring functions on the individual target in the DEKOIS2.0 data set; BEDROC ($\alpha = 80.5$) values of scoring functions on the individual target in the DEKOIS2.0 data set; EF_{0.5%} values of scoring functions on the individual target in the DEKOIS2.0 data set; EF_{5%} values of scoring functions on the individual target in the DEKOIS2.0 data set; and AUROC, BEDROC ($\alpha = 80.5$), EF_{0.5%}, EF_{1%}, and EF_{5%} values of scoring functions on the individual target in the DUD-E data set (PDF)

Various performance indicators for each target in the DEKOIS2.0 data set (XLSX)

Various performance indicators for each target in the DUD-E data set (XLSX)

Accession Codes

The PDBbind data set and CASF-2016 benchmark are available at <http://www.pdbbind.org.cn>. The cross-docking data set and the DEKOIS2.0/DUD-E data set can be found at <https://zenodo.org/record/5525936> and <https://zenodo.org/record/6623202>, respectively. The codes are available at <https://github.com/sc8668/RTMScore>.

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■ ABBREVIATIONS USED

AI, artificial intelligence; AR, androgen receptor; AUROC, area under the receiver operating characteristic curve; BEDROC, Boltzmann enhanced discrimination of receiver operating characteristic; CASF, Comparative Assessment of Scoring Functions; DEKOIS, Demanding Evaluation Kits for Objective in Silico Screening; DL, deep learning; DUD, Directory of Useful Decoys; DUD-E, Database of Useful Decoys-Enhanced; EF, enrichment factor; EM, cryo-electron microscopy; MDN, mixture density network; MHA, multihead self-attention; ML, machine learning; MLSE, machine-learning-based scoring function; MM/GBSA, molecular mechanics/generalized Born surface area; NMR, nuclear magnetic resonance; RF, random forest; RMSD, root-mean-square error; R_p, Pearson correlation coefficient; R_s, Spearman correlation coefficient; SF, scoring function; SR, success rate; SP, standard precision; SVM, support

vector machine; VS, virtual screening; 2D, two-dimensional; 3D, three-dimensional

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